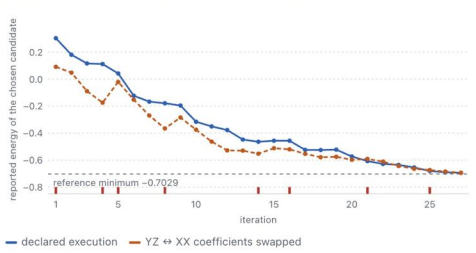
(a)

PART 0

Why verify? A wrong energy can look right



A VQE run is usually judged by its energy curve: if it falls steadily toward a minimum, the run looks healthy. The chart takes the measured counts from one declared configuration with two runs on hardware: once with the declared Hamiltonian coefficients, and once with the YZ and XX coefficients swapped.

Both curves fall smoothly. Nothing in the plot shows that the swapped version uses the wrong operator, yet it **would choose a different candidate in 8** of the 27 iterations.

The paper injects this exact fault (`coefficient_swap`) into real executions. The numeric check N2, which rebuilds each energy from the declared coefficients, reports DISAGREEMENT in 10/10 runs. The temporal replay (DejaVu) accepts 10/10, because every step still happens in the right order.

(b)

Choose a change

Original model
No change. All 24 properties hold.

REWIRE AN ARROW

Measure without rebinding
After one candidate, go straight to measurement without binding the next candidate's parameters.
`M5 · targets L3 · breaks 2`

Budget exhaustion aborts
When the budget is spent, go to ABORT instead of DONE.
`M10 · targets L4 · breaks 3`

Keep going past the budget
When the budget is spent, start another iteration instead of stopping.
`M12 · targets L4 · breaks 7`

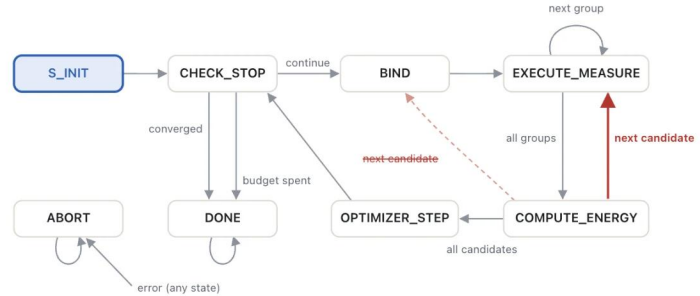
CHANGE OR REMOVE A GUARD

Aggregate after one group
The measure loop exits after the first group instead of all three.
`M1 · targets L1 · breaks 1`

Commit after one candidate
The optimizer steps after validating one candidate instead of three.
`M2 · targets L2 · breaks 3`

Accept success only at the budget
Convergence leads to DONE only at the last iteration; earlier success falls through to ABORT.
`M11 · targets L4 · breaks 3`

Ignore measurement failures
measure_fails is removed from the error signal, so a failed measurement does not abort.
`M9 · targets L4 · breaks 4`



THE EDIT TO THE NUXMV MODEL

```
- state = COMPUTE_ENERGY & cands_done + 1 < NCANDS : BIND;
+ state = COMPUTE_ENERGY & cands_done + 1 < NCANDS : EXECUTE_MEASURE;
```

Red marks the change on the diagram. Struck-out dashed arrows can no longer fire.

Properties

2 fail · 22 still hold · detected by contract L3

READINESS
- R1 groups before aggregation
- R2 energy before check
- R3 bound before measure
- R4 candidates before commit

FRESHNESS
- F1 current evaluation
- F2 current energy
- F3 evaluation reset
- F4 iteration reset
- F5 returned energy

TERMINATION
- T1 termination
- T2 done reason
- T3 done absorbing
- T4 abort absorbing
- T5 success reason
- T6 budget reason
- T7 abort reason
- T8 no early reason
- T9 update accounting

FAILURE HANDLING
- E1 bind fails
- E2 measure fails
- E3 energy fails
- E4 optimizer fails
- E5 abort only on error
- E6 update bound

(c)

One complete run showing the parameter and energy updates

27 iterations · stopped by success

ENERGY UPDATES BY ITERATION

▶ Play run

